

Population-scalable Genotyping from Low-coverage Sequencing using Pangenome Graphs

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PANGENOME-BASED GENOTYPING OF STRUCTURAL VARIANTS IN MEDICAL COHORTS



Chiara Paleni

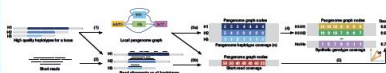
Chiara Paleni (1), Davide Bolognini (1), Andrea Guarracino (2), Thomas Sumner Dudley (1), Federica Santonastaso (1), Edoardo Giacomuzzi (1), Alessandro Raveane (1), Peter H Sudmant (3), Erik Garrison (2), Nicole Soranzo (1, 4, 5)

1. Background

- Pangenome graphs built from long-read assemblies enable comprehensive variant discovery at highly complex and structurally variable loci.
- Most large-scale biobanks rely on short-read sequencing; this limits pangenome potential for population and medical genomics applications.
- We developed COSIGT (COsine Similarity-based GenoTyper), a method which exploits short-read alignments and high-quality pangenomes to assign structural haplotypes to samples.
- We proved COSIGT has high accuracy at even low coverage levels (see bioRxiv preprint).
- We present an application of COSIGT and the HPRC Human Pangenome Reference Consortium for structural variant (SV) genotyping in a Southern-Italian medical cohort, representative of an understudied Mediterranean population: the Moli-sani cohort, which includes 14,000 participants with 15-years follow-up, extensive biomedical data, microarray genotyping, and short-read whole-genome sequencing (WGS) data.

2. COSIGT haplotype assignment

COSIGT assigns haplotype sequences from pangenome graphs to short-read samples by comparing read coverage and haplotype copy-number vectors. (1) Haplotype sequences for a locus of interest are aligned with Minimap2 and a local pangenome graph is built with PGGB. (2) Short reads aligned to the locus of interest are re-mapped against all haplotypes. (3a,b) Haplotype graph paths and read alignments are converted to coverage vectors in the node space. (4) All possible genotype vectors are computed by summing pairs of haplotype coverage vectors. (5) The sample and genotype coverage vectors are compared with cosine similarity and the pair of haplotypes with the highest score is selected as the sample genotype.



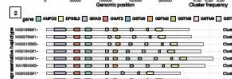
3. Pangenomes reveal structural variation

Haplotype node coverage vectors are grouped with density-based clustering, which allows identification of structural clusters representing similar large-scale structures. Clusters presenting large differences from the reference can have high frequencies in the cohort; this highlights the limits of single reference-based approaches.

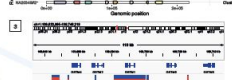
(1) The node coverage heatmap gives an overview of the haplotype structures in this locus. Grey nodes have coverage 1, colors represent copy-number variable regions; nodes in white represent deletions.



(2) Copy-number variation in protein-coding genes can be identified in the Pangenome visualization. Clusters 1 and 3 present GSTM1 deletion while Cluster 2 has a duplication.

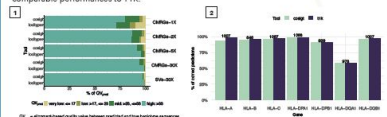


(3) Haplotypes are analysed with SVIM-asm to call structural variants relative to the hg38 reference. The genotypes of the 2 predicted haplotypes are then assigned to each sample.



4. COSIGT is accurate even at low coverage

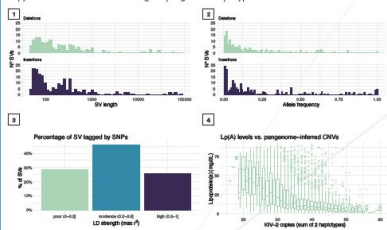
- We benchmarked COSIGT's performance on 327 loci including challenging, medically relevant genes (CMRGs) as well as 247 SV-enriched regions from HPRC) on publicly available genome assemblies and WGS datasets. We outperform Loctyper in low coverage scenarios.
- We tested COSIGT for HLA typing on a subset of the Moli-sani cohort (1005 samples) against imputed HLA-dig1 typing from microarray data and compared prediction accuracy to a specific WGS-based HLA typing software, TIK. Despite being a general-purpose tool, COSIGT reaches comparable performances to TIK.



5. Structural diversity accessible with short reads

We genotyped 44 loci (average length: 148bp), selected from the SV and CMRG list (see panel 4) and containing genes relevant to cardiovascular disease, in the Moli-sani cohort. We identified 455 SVs, of which 357 were present with a minor allele frequency (MAF) $\geq 1\%$ in the cohort.

- Length and frequency of SVs with MAF $\geq 1\%$ in the Moli-sani cohort. Most SVs are below 500 bp.
- Linkage disequilibrium (LD) between SVs and SNPs called on the same WGS data. A large portion of SVs are not in strong LD with a SNP ($r^2 < 0.8$).
- The LPA locus harbors a well-known association between the number of copies of the KIV-2 (kringle IV-2) repeat region and levels of lipoprotein A. This association can be recovered in the Moli-sani with copy numbers inferred from the assigned pangenome haplotypes.



6. Open challenges

- SVIM-asm cannot call multi-allelic CNVs that we can see from the node coverage plots. Can we call variants directly from the graph?
- Pangenome-based SVs open areas of the genome inaccessible to SNPs to association studies. Can we find association models which integrate all types of SVs?
- Are there SVs in linkage with SNPs associated to certain phenotypes? (Potential causal variants)
- How to assess the confidence of individual variant calls beyond cosine similarity?
- Application to UK Biobank and integration with molecular phenotypes (gene expression, cell type composition...)

YOU HAVEN'T MISSED THIS, RIGHT?!? YOU STILL HAVE TIME!

Chiara Paleni
PostDoc @ Human Technopole



Genotyping, SV Calling and Association Testing using Pangenomes Graphs

Davide Bolognini

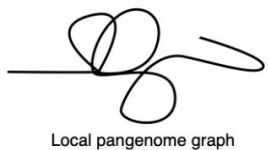
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H1
H2
H3
High quality haplotypes for a locus

Short reads

(1)
(2)



H1
2
3
Read alignments on all haplotypes

(3a)
(3b)

Pangenome graph nodes

H1	5	5	5	4	4	4	4		
H2	5	4	5	3	2	2	3		
H3	1	1	2	1	1	1	1		

Pangenome haplotype coverage (n)

Pangenome graph nodes

	50	50	40	40	40	40	35		
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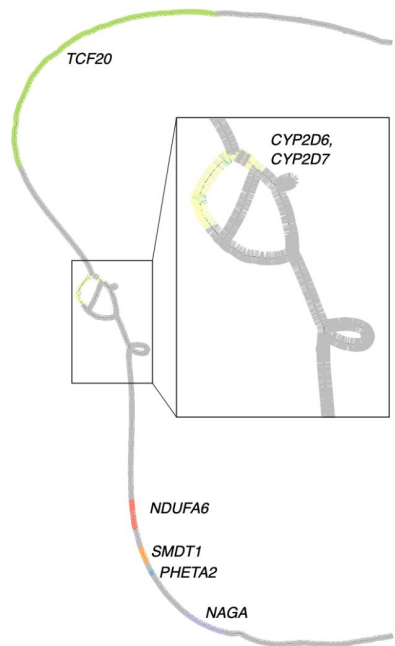
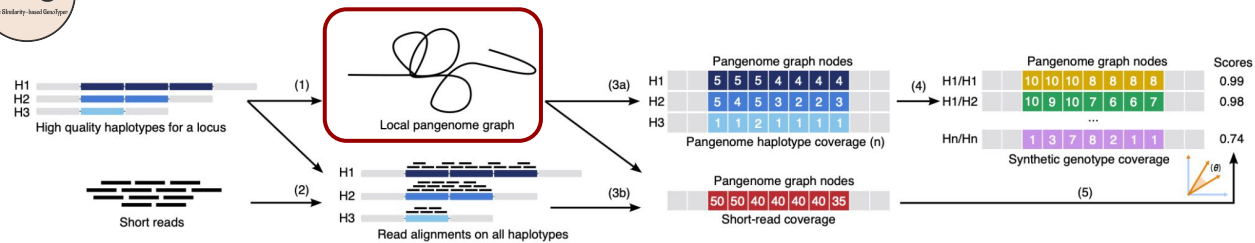
Short-read coverage

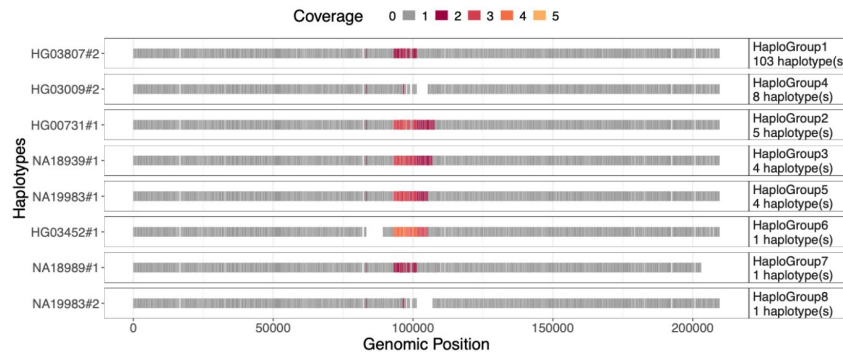
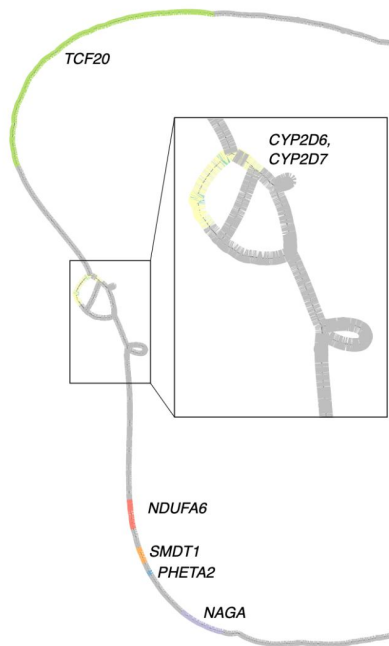
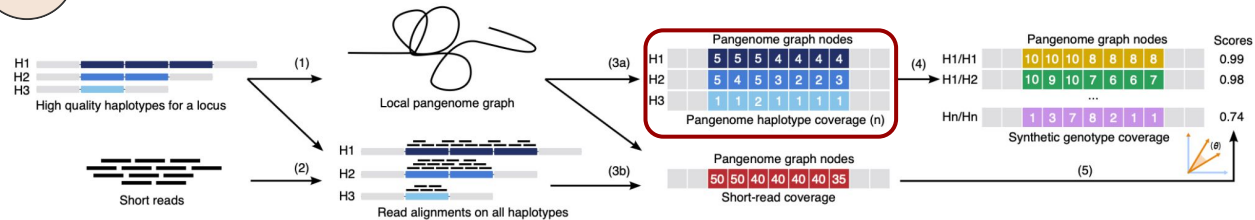
(4)

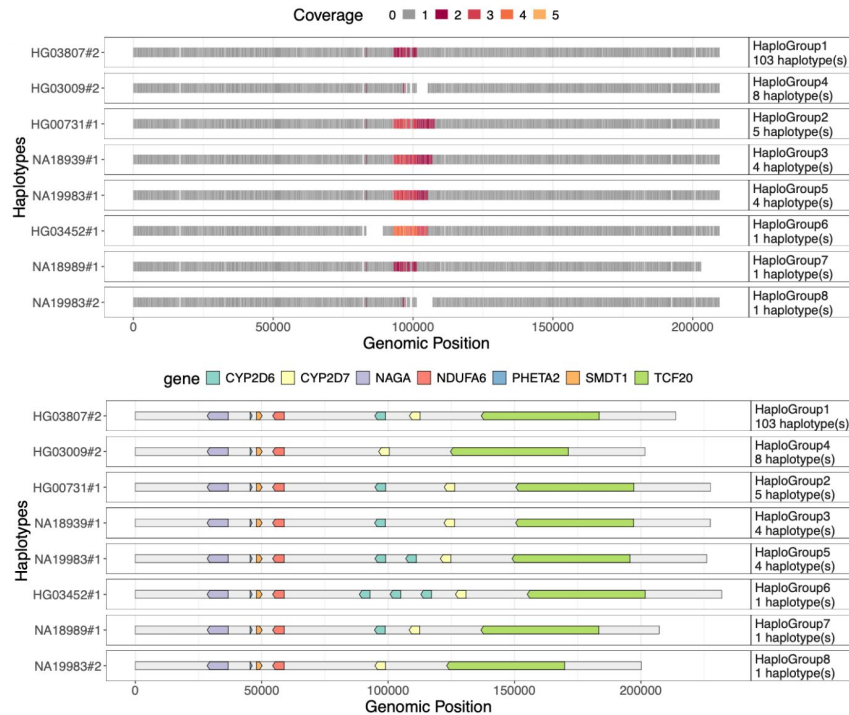
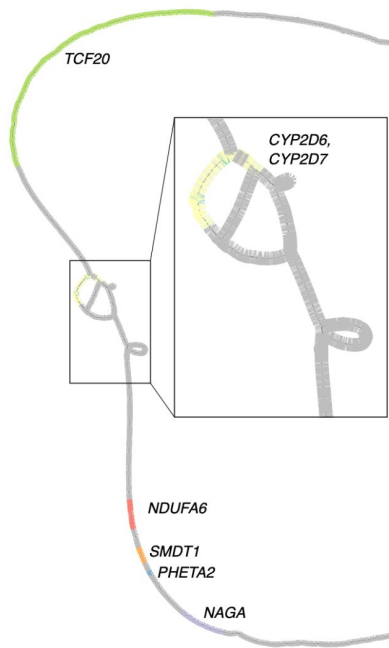
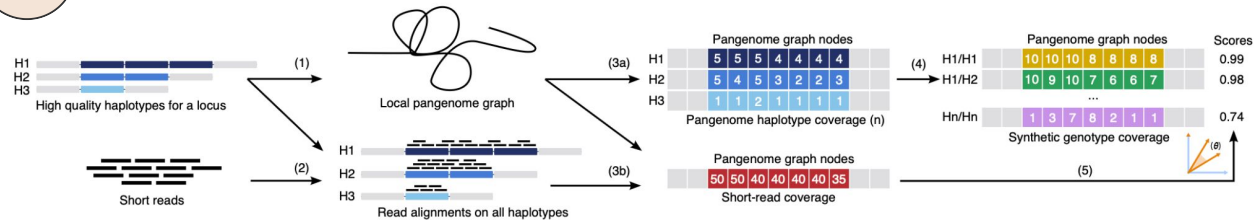
H1/H1	10	10	10	8	8	8	8			Scores	0.99
H1/H2	10	9	10	7	6	6	7				0.98
Hn/Hn	1	3	7	8	2	1	1				0.74

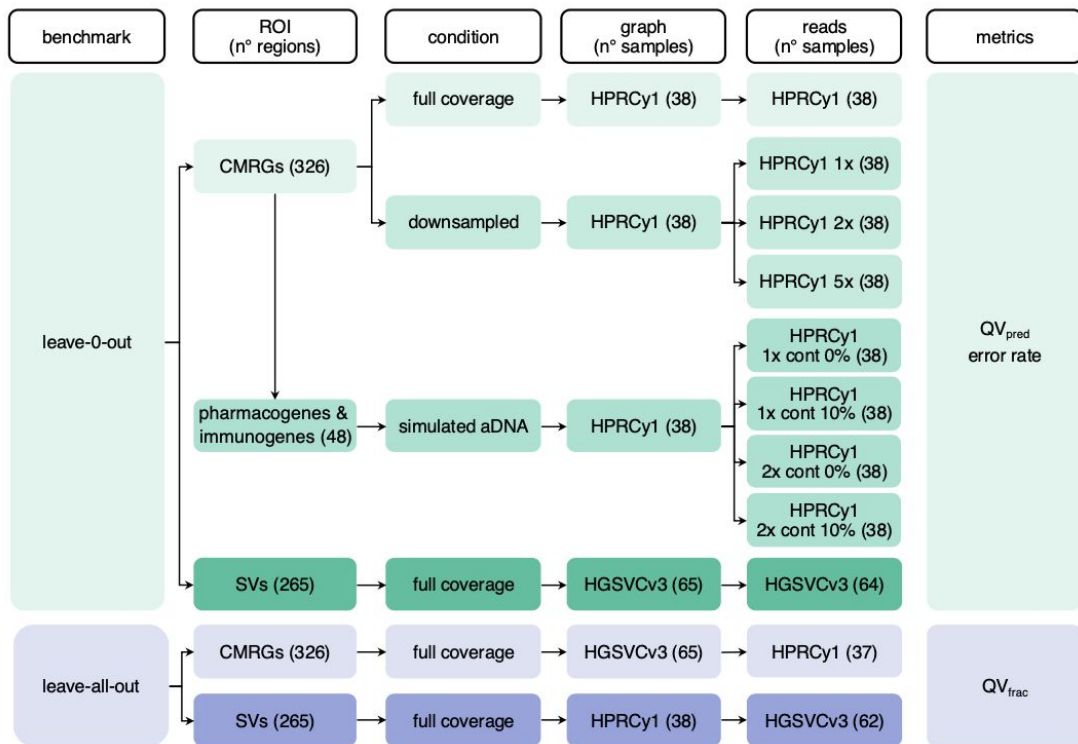
Synthetic genotype coverage

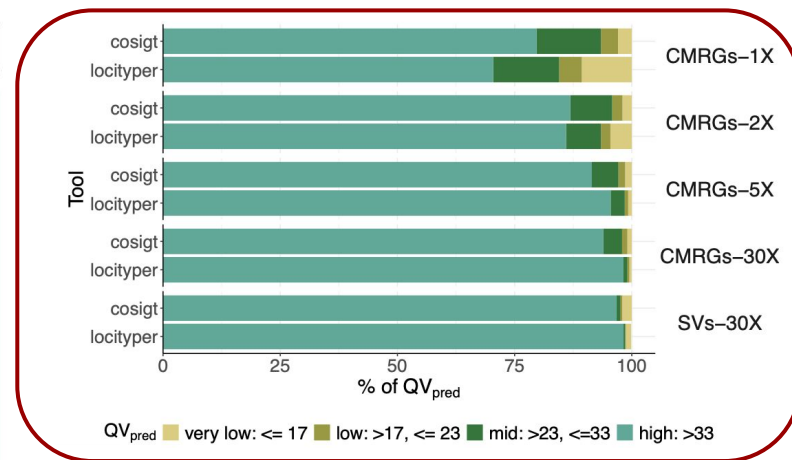
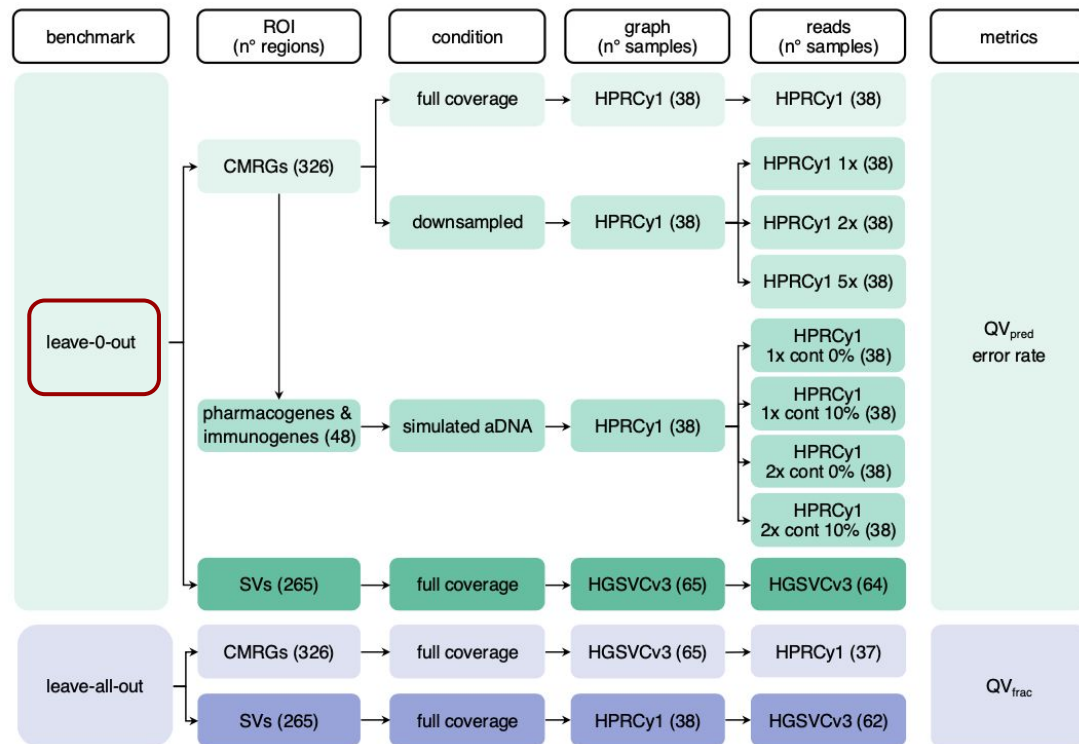
(5)

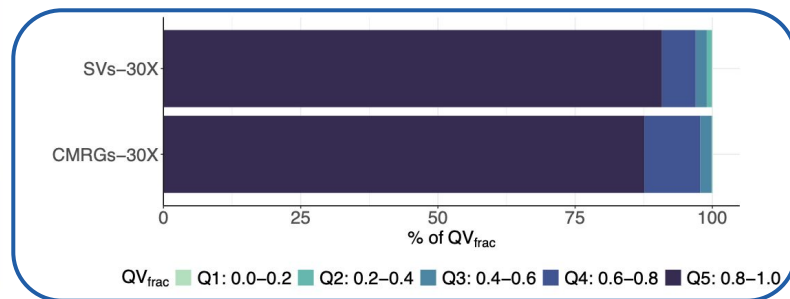
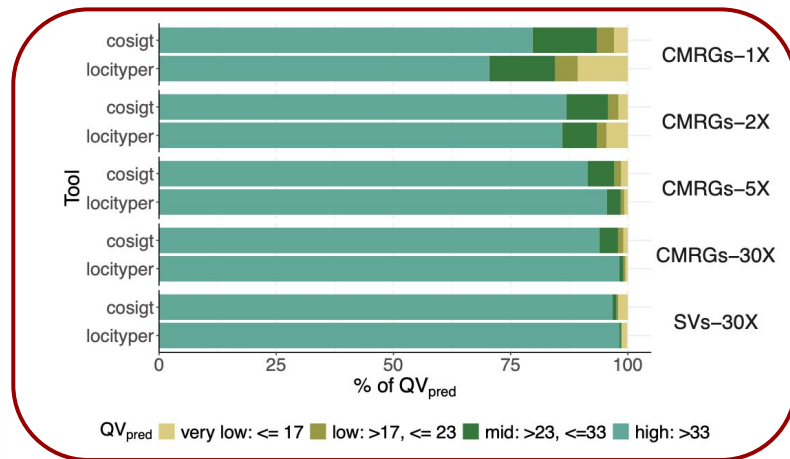
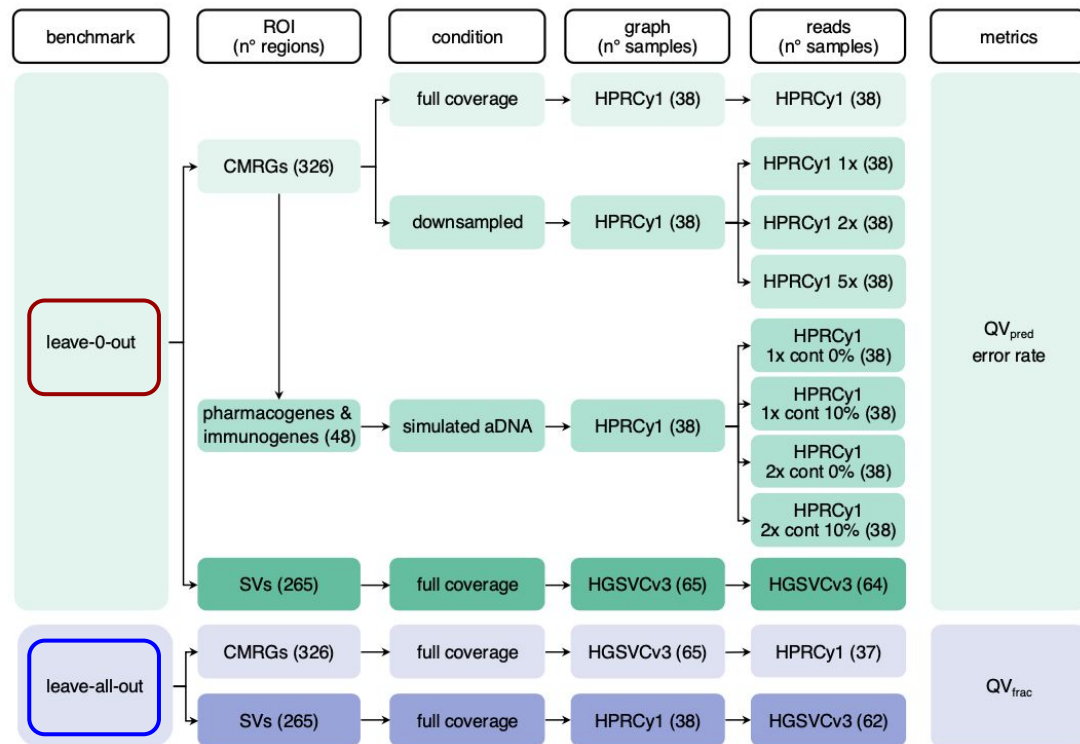


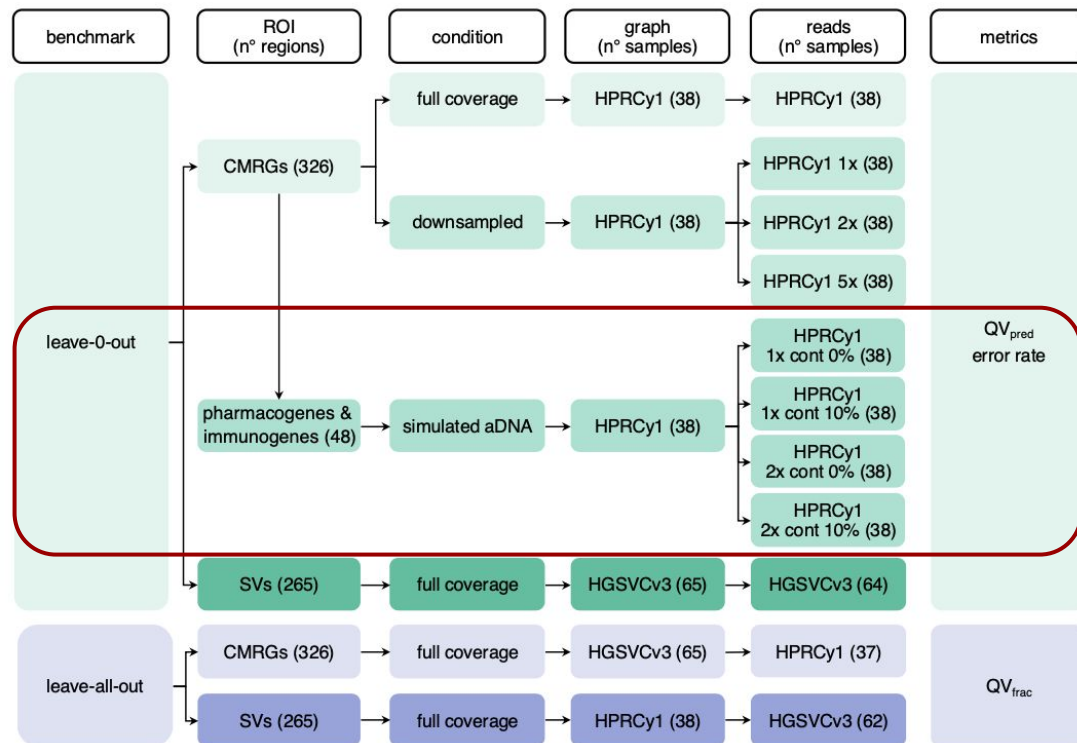


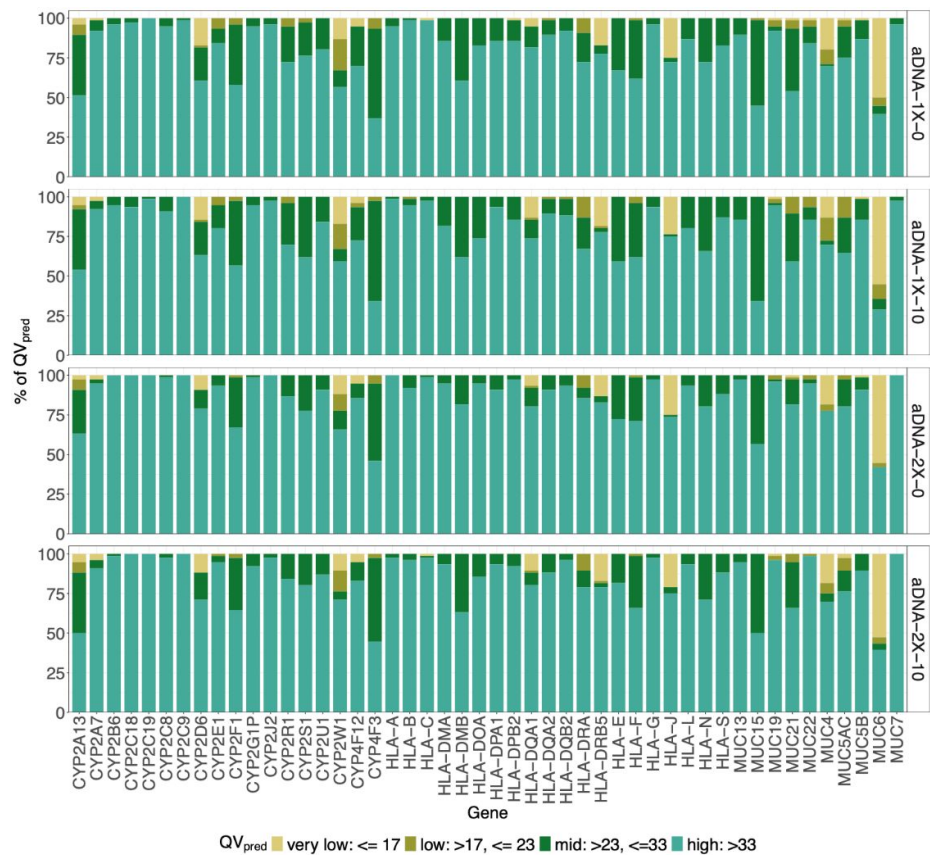


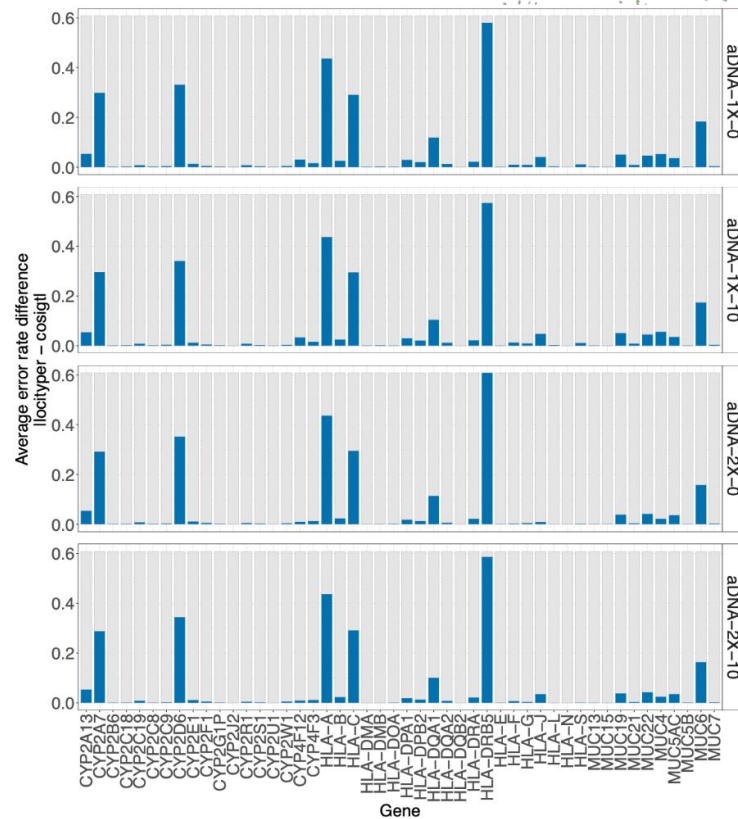
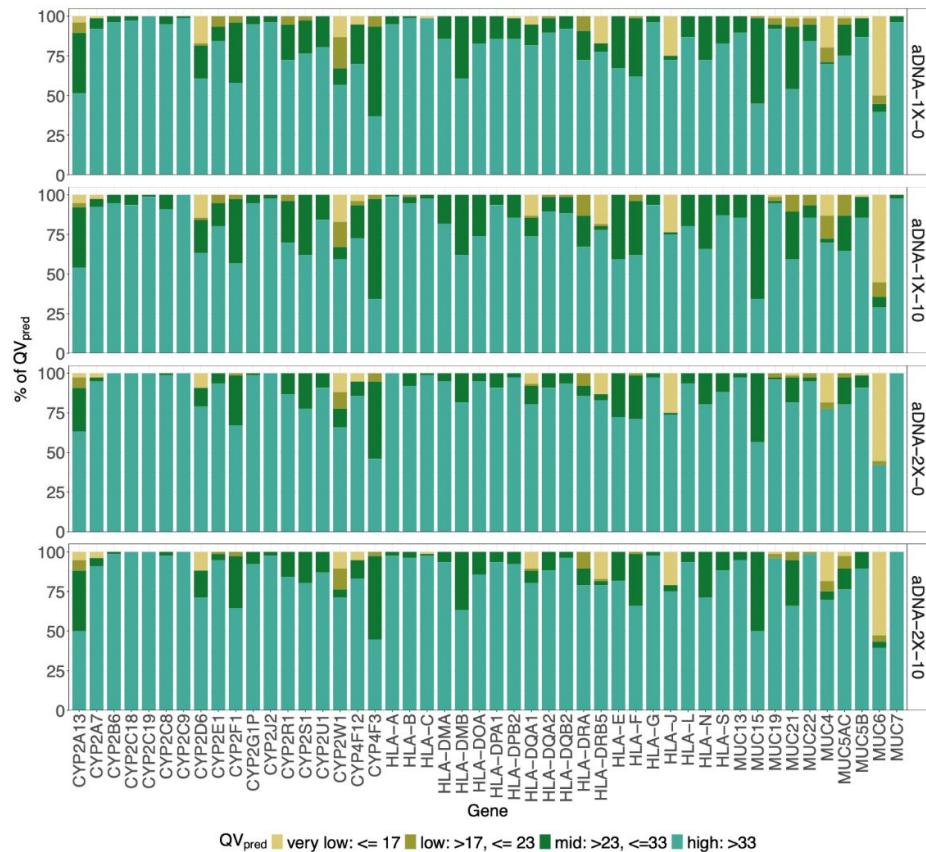


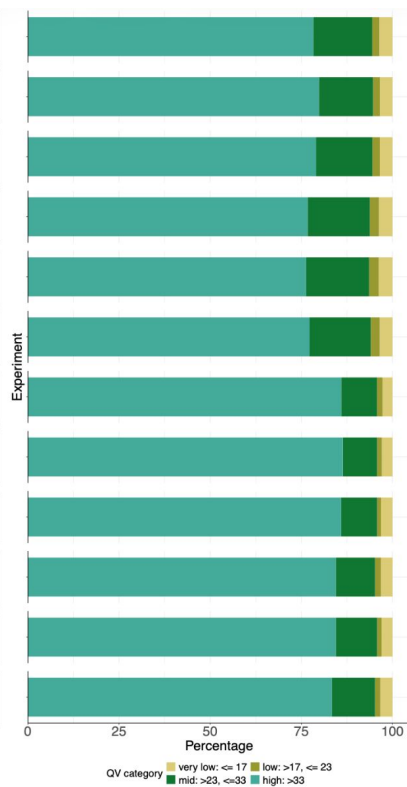
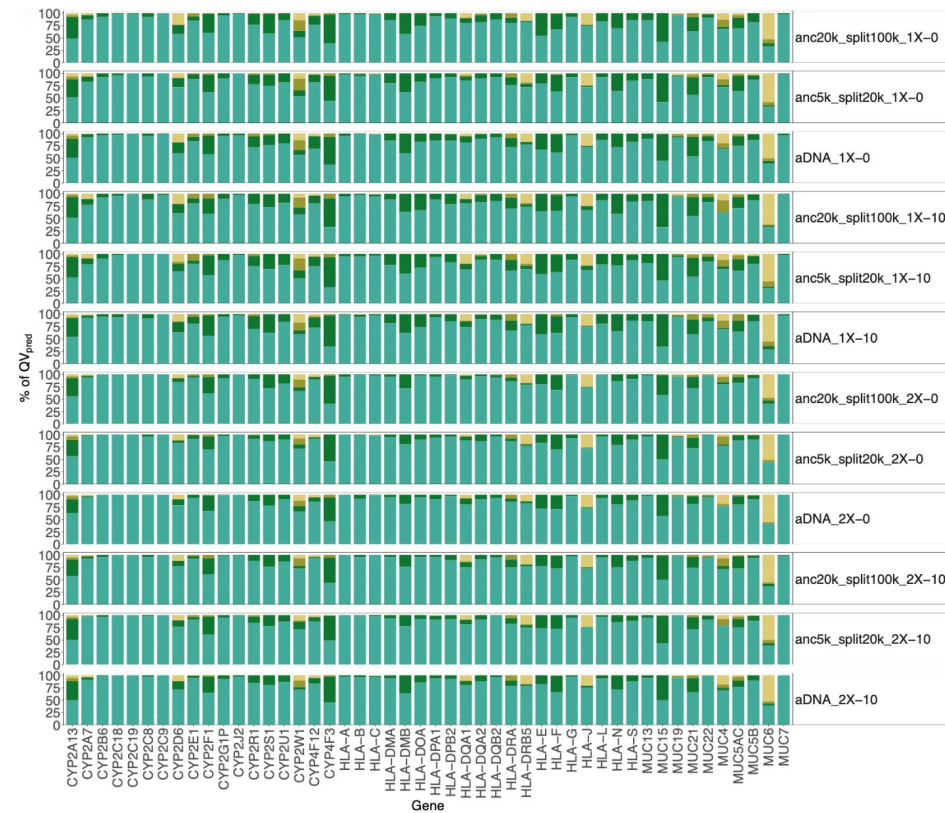


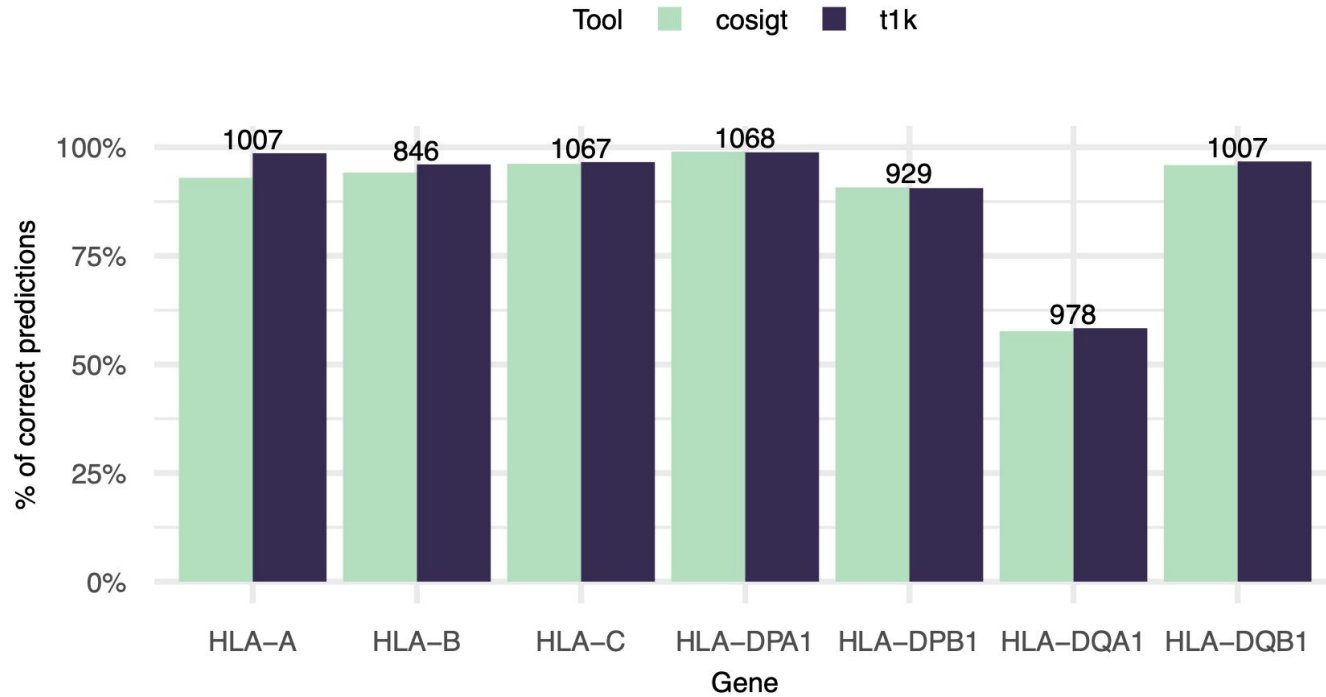


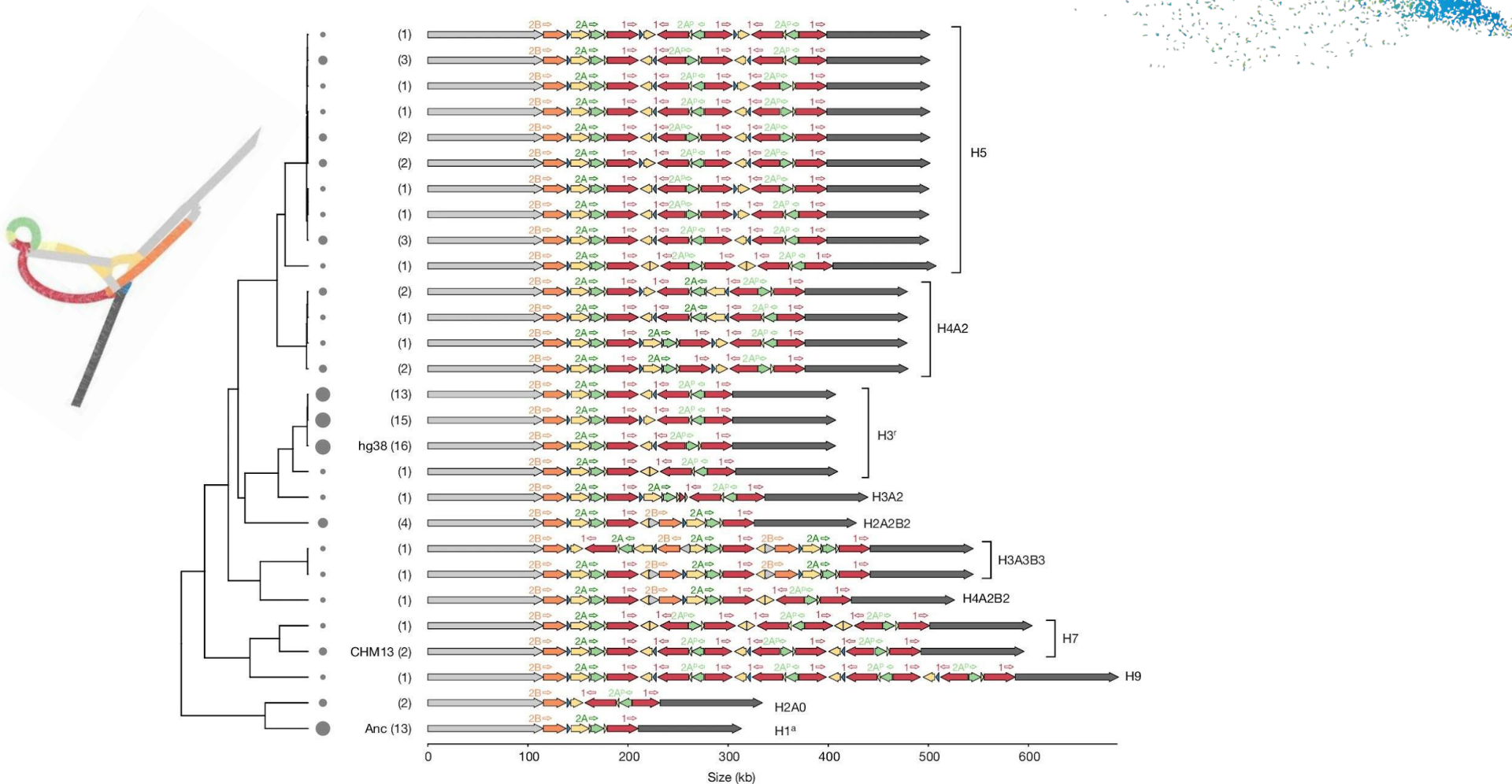


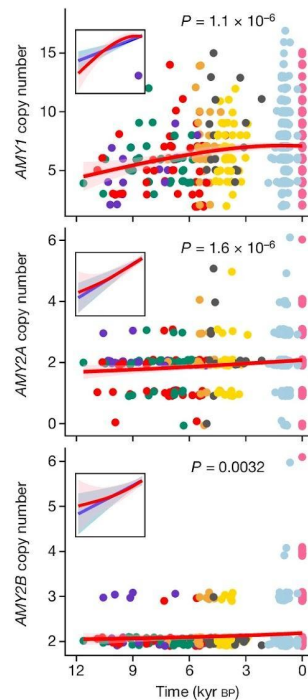
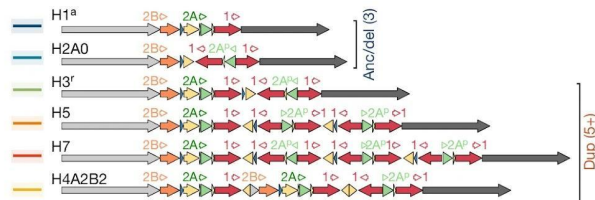
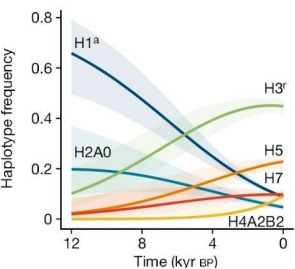










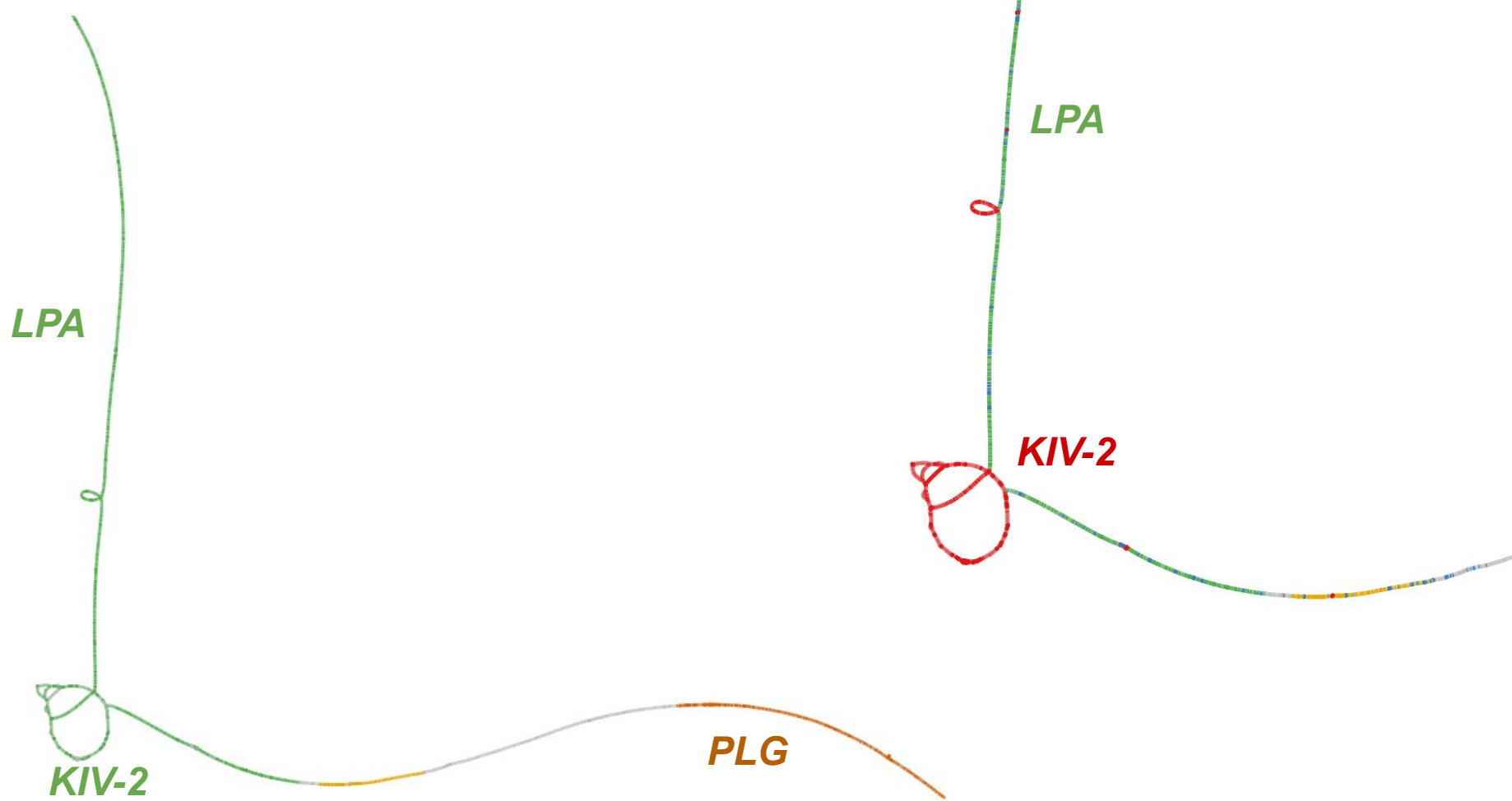


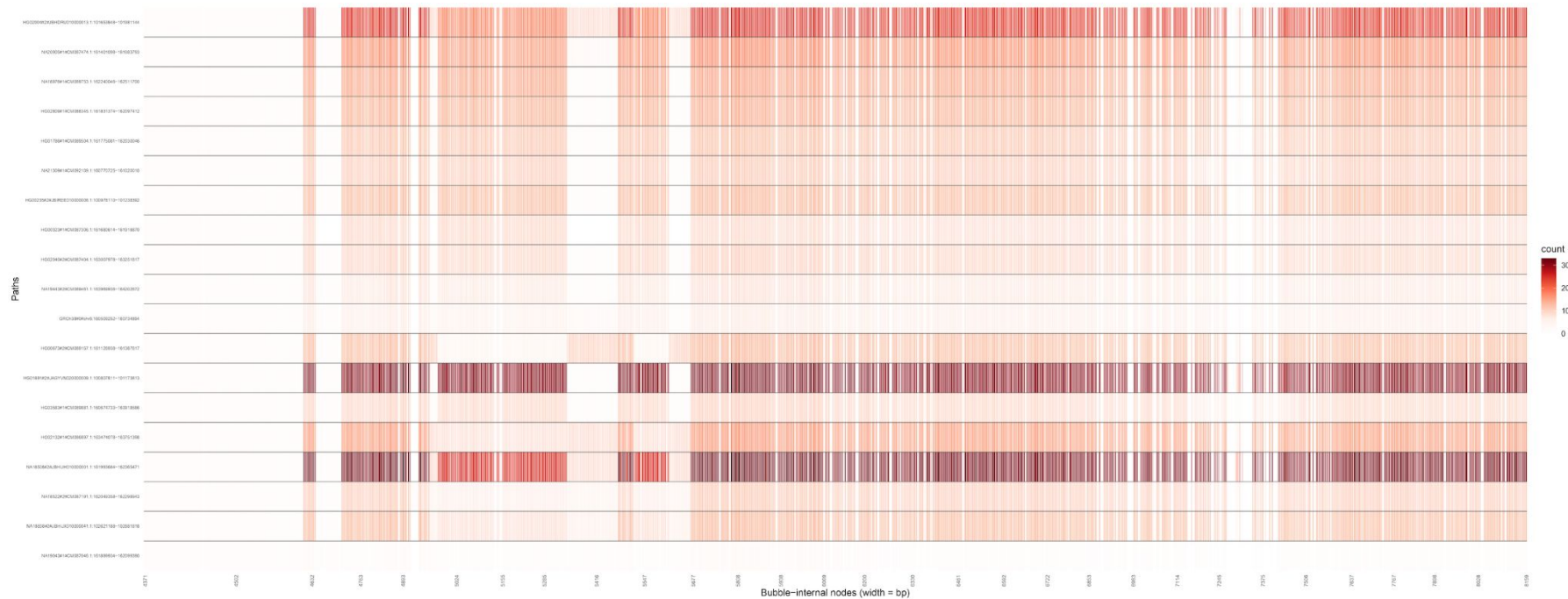
LPA

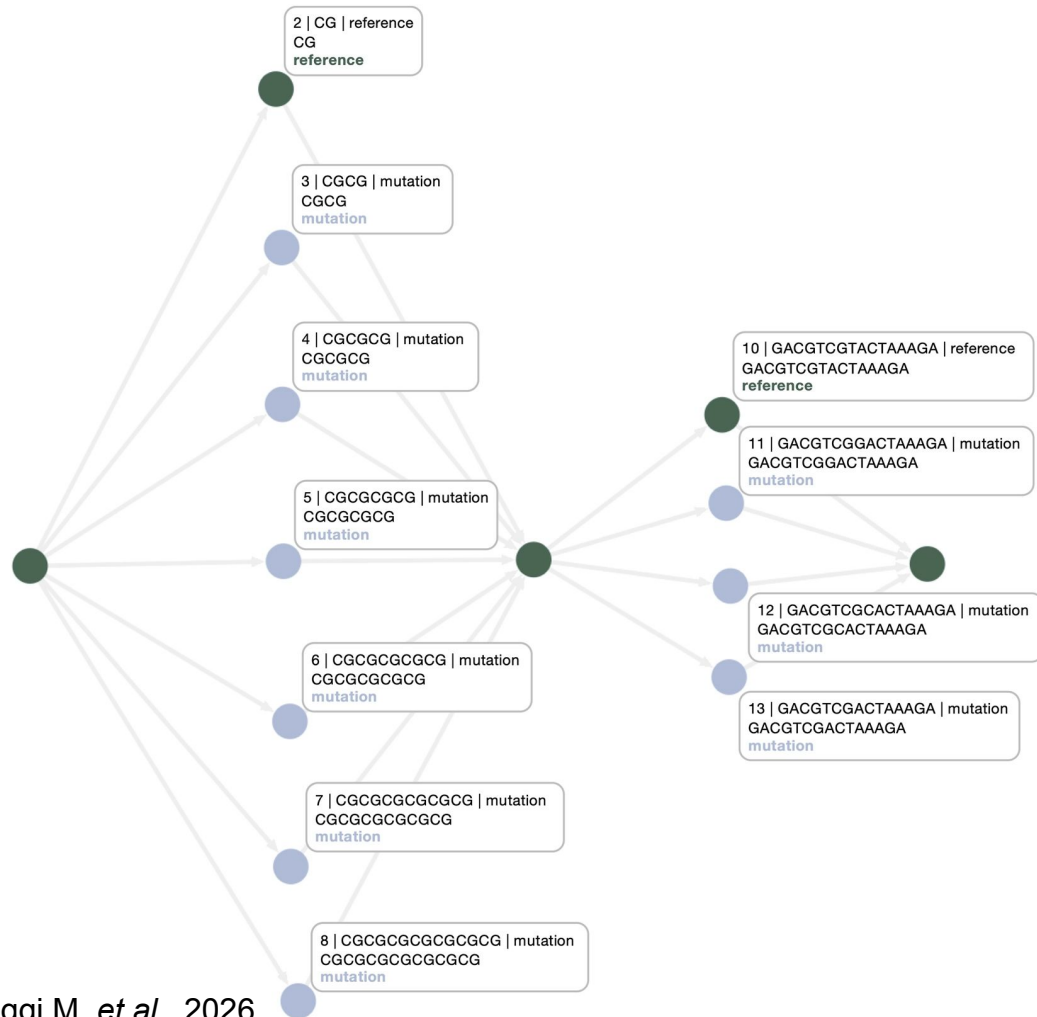


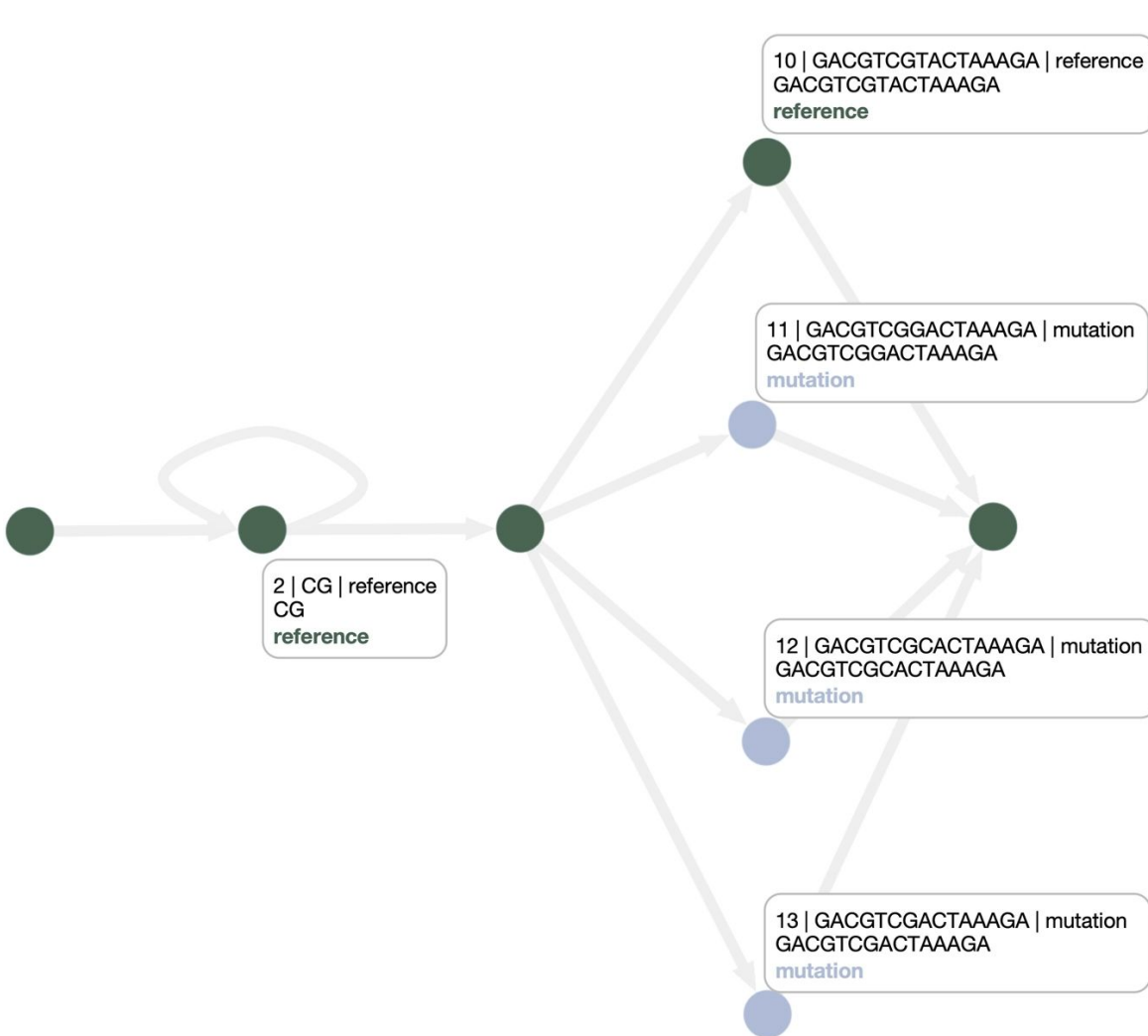
KIV-2

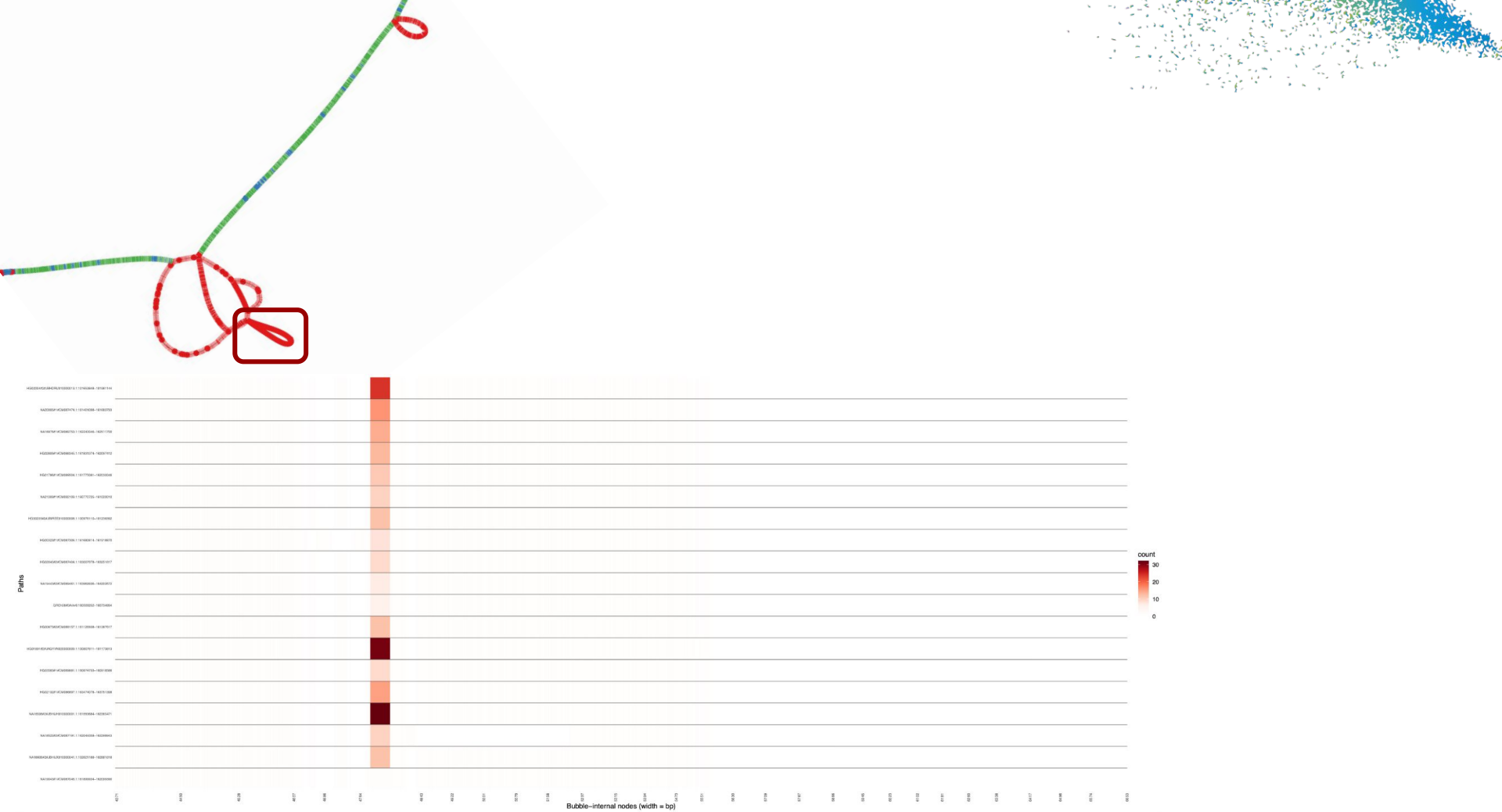
PLG

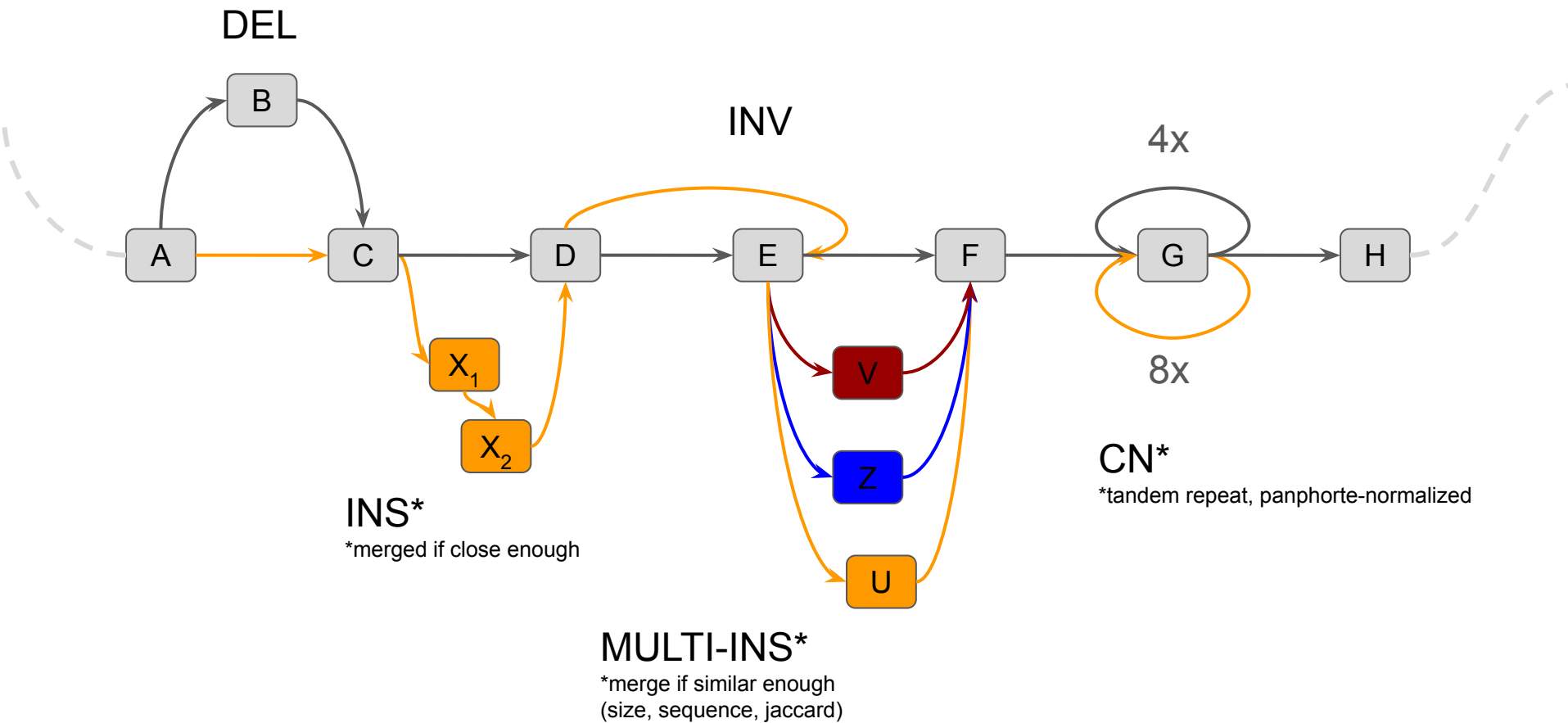


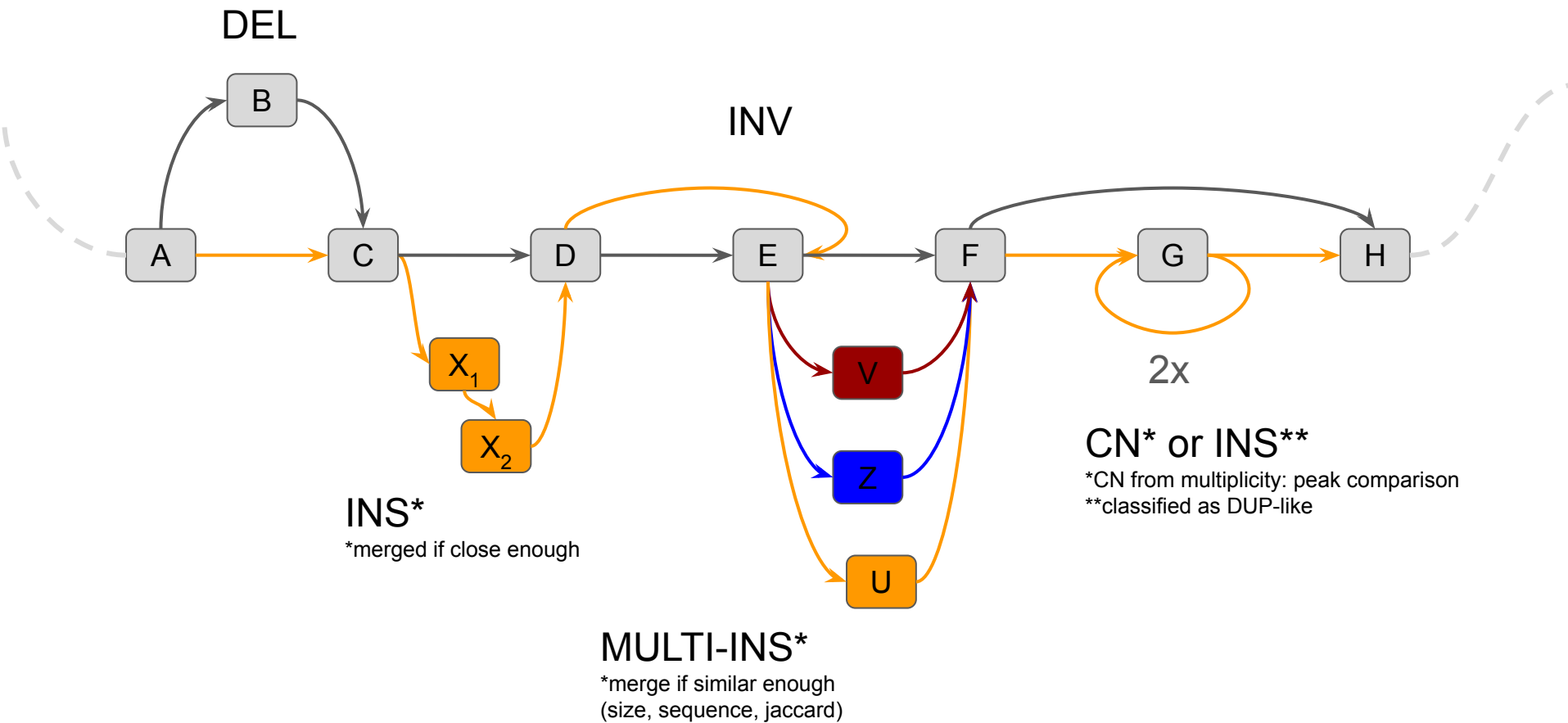


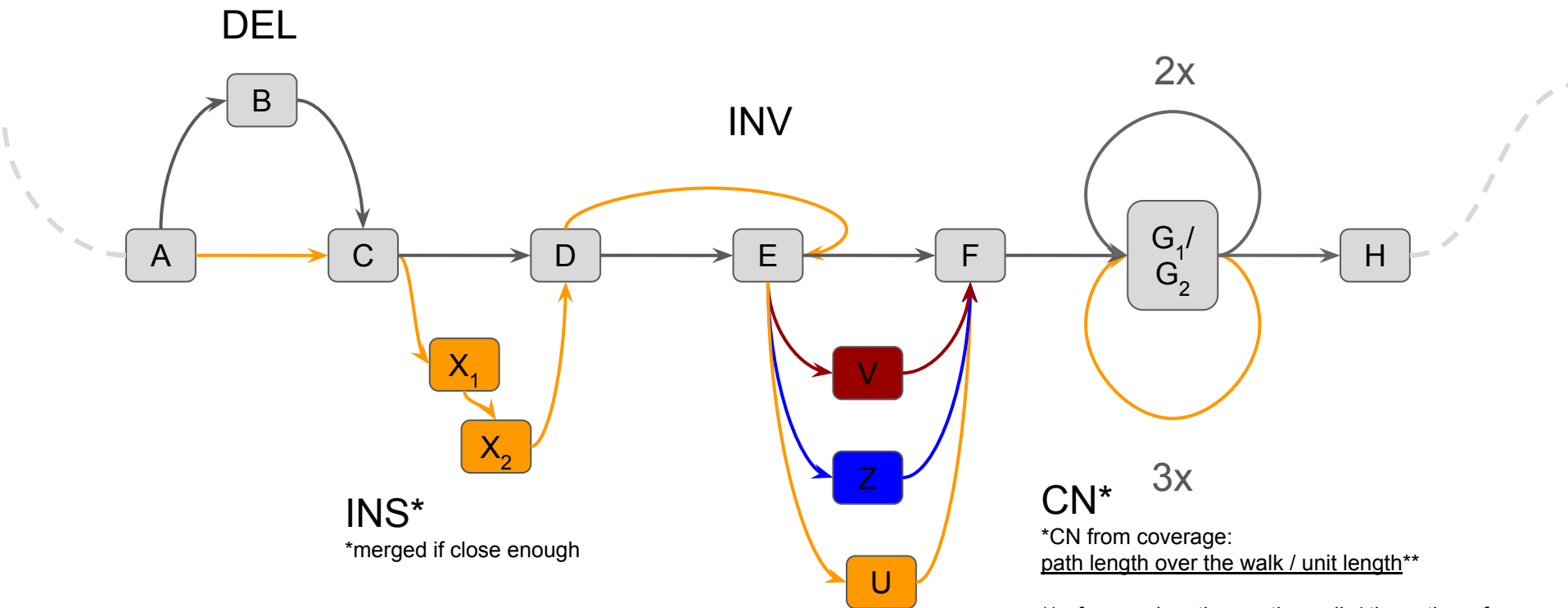












INS*

*merged if close enough

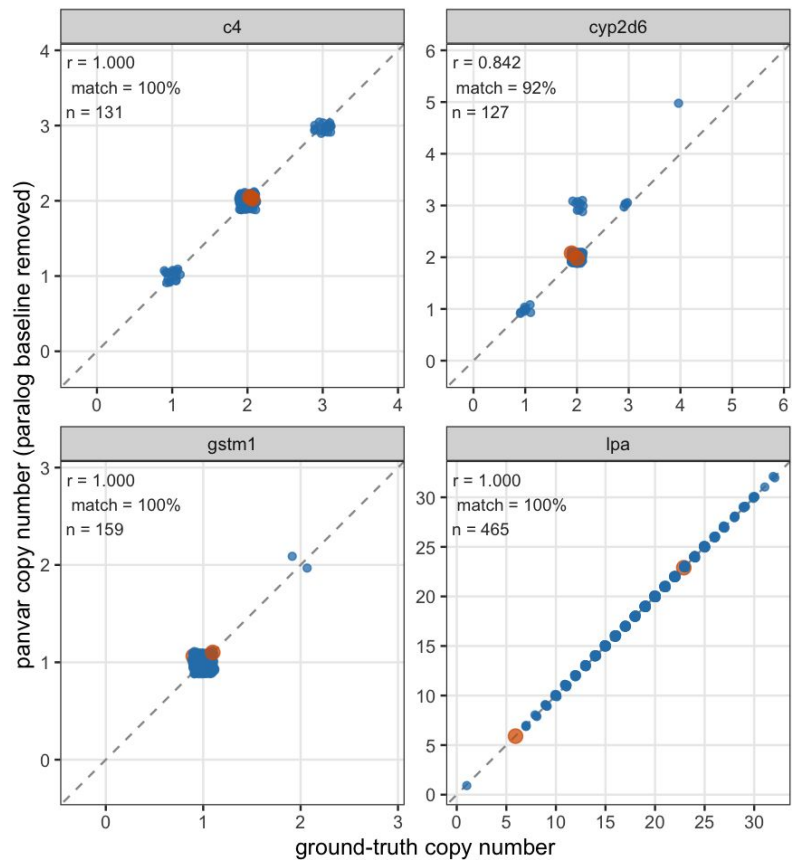
MULTI-INS*

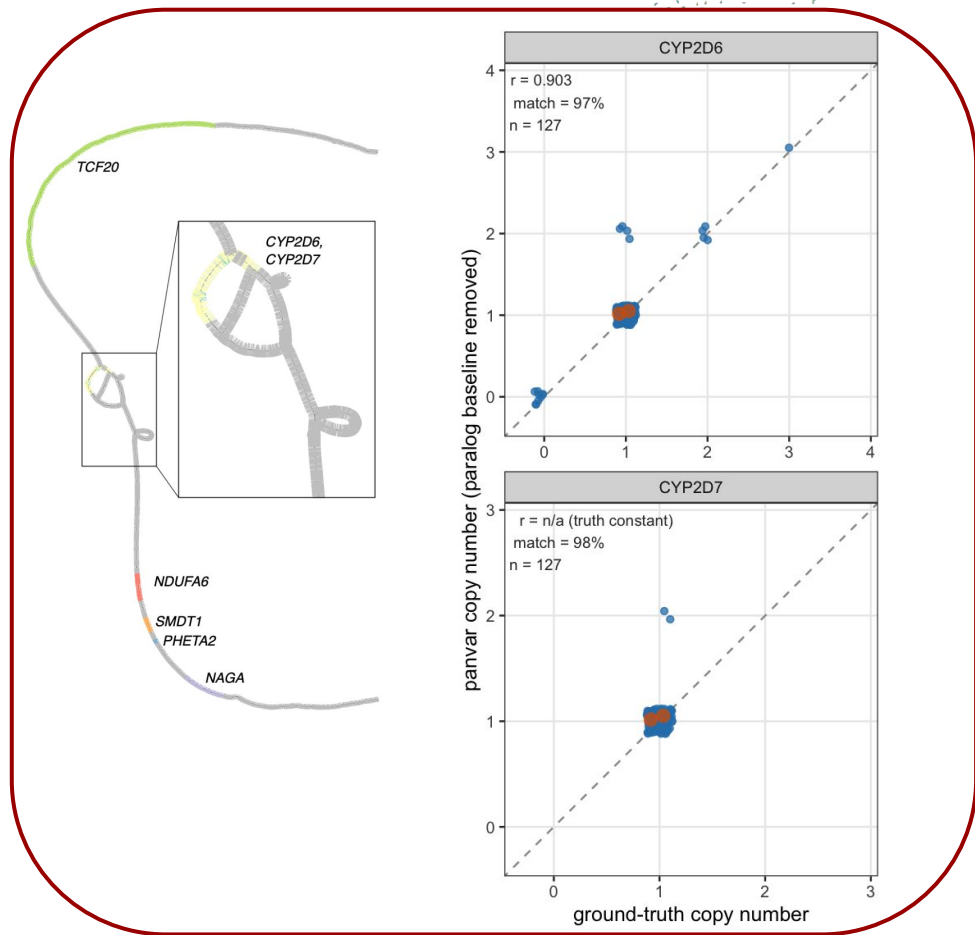
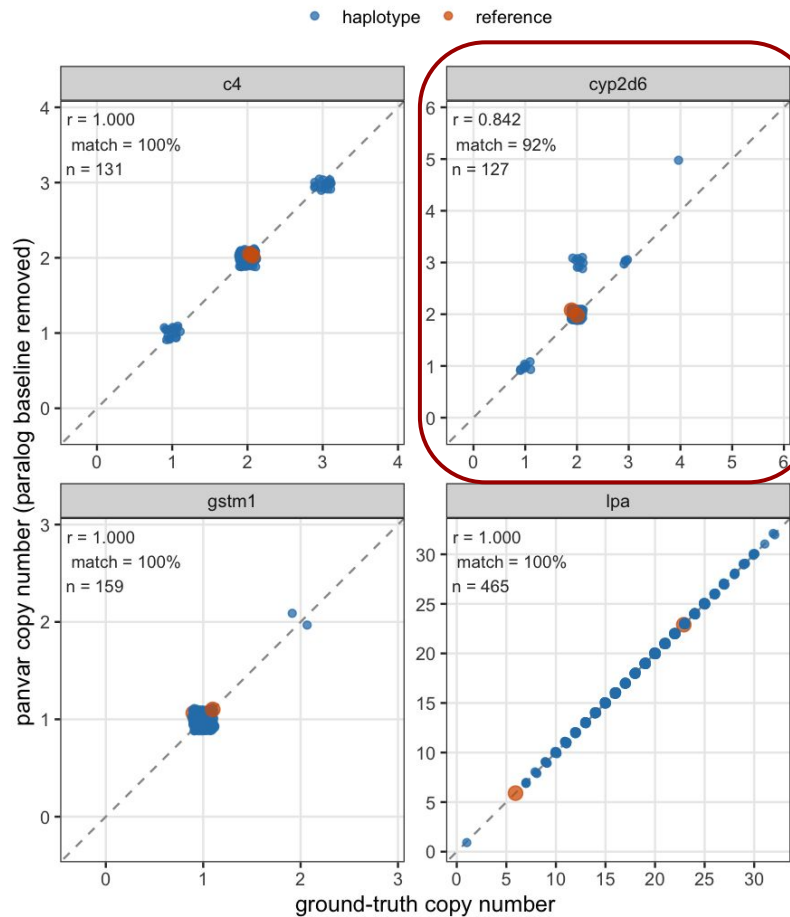
*merge if similar enough
(size, sequence, jaccard)

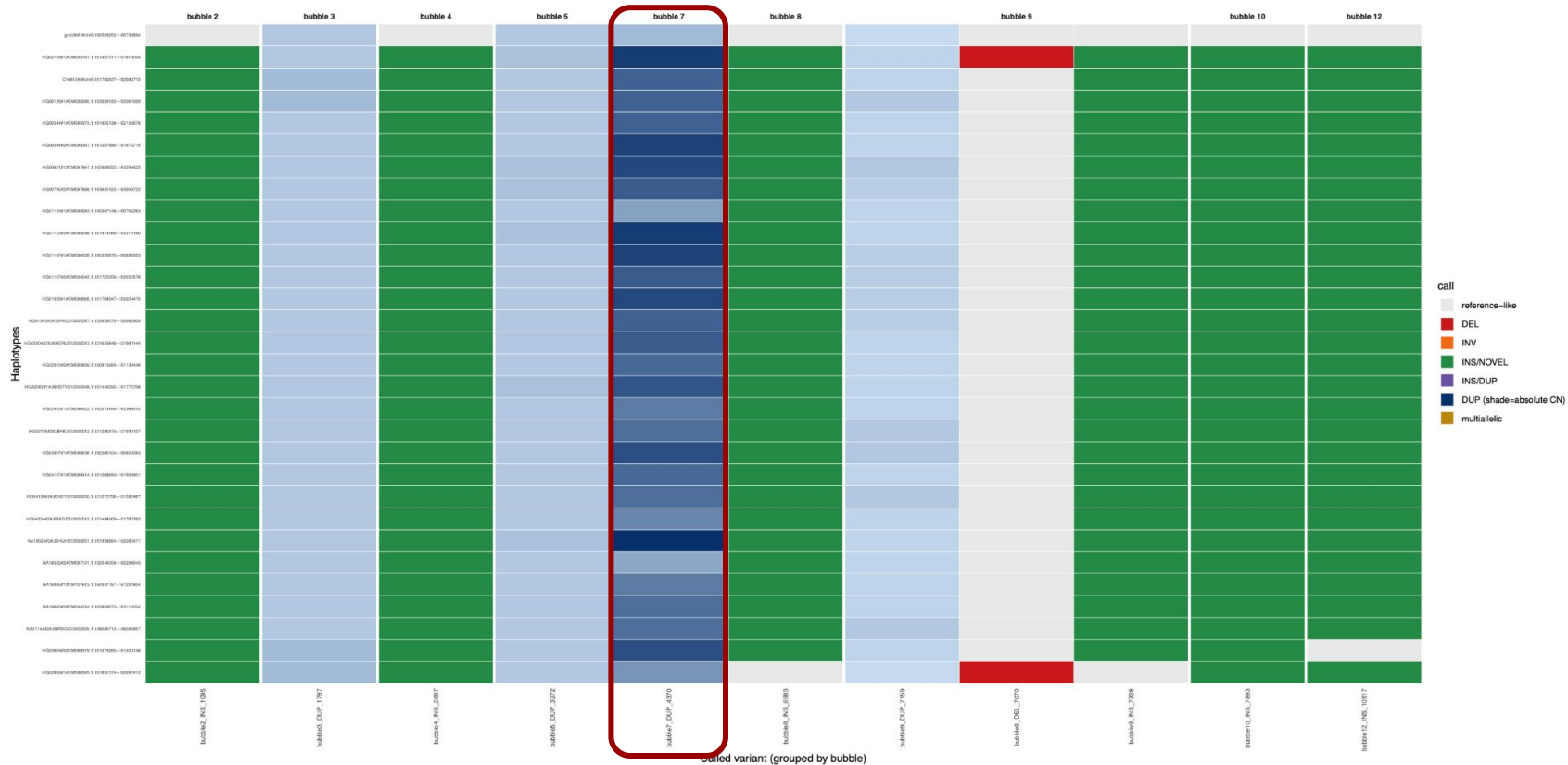
*CN from coverage:
path length over the walk / unit length**

**reference length over the walk / times the reference fold

● haplotype ● reference









A team effort!

HT, PopMed



TGEN



UTHSC



UCB



UNIPV

NEUROMED

COSIGT: the tooling ecosystem



1. <https://github.com/davidebolo1993/cosigt>: end-to-end genotyping pipeline.
 - a. **master**: whole-chromosome assemblies to genotypes (standard short reads)
 - b. **ancient_dna**: whole-chromosome assemblies to genotypes (aDNA short reads)
 - c. **custom_alleles**: locus-specific assemblies to genotypes (standard short reads)
 - d. **unify**: whole-chromosome/locus specific assemblies to genotypes (standard and aDNA reads, long reads) → will be merged into **master** soon, unifying the different branches
2. <https://github.com/pangenome/impvg>: implicit pangenome graphs
3. <https://github.com/davidebolo1993/kfilt>: recruit unmapped reads to pangenome alleles
4. <https://github.com/AndreaGuarracino/panplexity>: low-complexity regions in pangenome graphs
5. <https://github.com/davidebolo1993/ancestralsim>: aDNA simulation from assemblies
6. <https://github.com/davidebolo1993/panvar>: SV calling/describing for associations (beta, private)
7. <https://github.com/GenoGra/Panphorte>: collapse tandem repeats in pangenome bubbles



Accessible pangenome analyses @ GenoGra



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pangenomics?
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Thank you!

